

Master CIDR Subnetting Reference Guide

Horizontal Overview Across All Four IPv4 Octets

This reference document maps out Classless Inter-Domain Routing (CIDR) prefixes from /1 to /32 horizontally across each octet. Use this chart to quickly determine subnet masks, total available subnets, and host capacities.

The Core Subnetting Formulas:

Total Subnets = $2^{\text{Bits Borrowed (n)}}$

Total Hosts = $2^{32 - \text{CIDR Prefix (h)}}$

Usable Hosts = *Total Hosts* - 2 (Subtracts Network ID & Broadcast ID)

1st Octet Subnetting: /1 to /8

Bits are borrowed from the first octet. These represent massive global network blocks.

CIDR Prefix	/1	/2	/3	/4	/5	/6	/7	/8 (Class A)
Subnet Mask	128.0.0.0	192.0.0.0	224.0.0.0	240.0.0.0	248.0.0.0	252.0.0.0	254.0.0.0	255.0.0.0
Total Subnets	2	4	8	16	32	64	128	1
Total Hosts	2,147,483,648	1,073,741,824	536,870,912	268,435,456	134,217,728	67,108,864	33,554,432	16,777,216

2nd Octet Subnetting: /9 to /16

Bits are borrowed from the second octet. Commonly used by large enterprises and ISPs.

CIDR Prefix	/9	/10	/11	/12	/13	/14	/15	/16 (Class B)
Subnet Mask	255.128.0.0	255.192.0.0	255.224.0.0	255.240.0.0	255.248.0.0	255.252.0.0	255.254.0.0	255.255.0.0
Total Subnets	2	4	8	16	32	64	128	1
Total Hosts	8,388,608	4,194,304	2,097,152	1,048,576	524,288	262,144	131,072	65,536

3rd Octet Subnetting: /17 to /24

Bits are borrowed from the third octet. Ideal for medium to large campus networks.

CIDR Prefix	/17	/18	/19	/20	/21	/22	/23	/24 (CI
Subnet Mask	255.255.128.0	255.255.192.0	255.255.224.0	255.255.240.0	255.255.248.0	255.255.252.0	255.255.254.0	255.25
Total Subnets	2	4	8	16	32	64	128	
Total Hosts	32,768	16,384	8,192	4,096	2,048	1,024	512	25

4th Octet Subnetting: /25 to /32

Bits are borrowed from the final octet. This is used for local area network (LAN) sizing and tight host control.

CIDR Prefix	/25	/26	/27	/28	/29	/30	/31	/32
Subnet Mask	...255.128	...255.192	...255.224	...255.240	...255.248	...255.252	...255.254	...255.255
Total Subnets	2	4	8	16	32	64	128	256
Total Hosts	128	64	32	16	8	4	2	1
Usable Hosts	126	62	30	14	6	2	2*	1*

* Note: /31 profiles are typically used for point-to-point links (RFC 3021). /32 designates a single dedicated host loopback address.

Key Structural Pattern for Students

Teach students to look for the repeating bit values across every single octet. The custom decimal value of the mask remains exactly identical depending on how many bits are borrowed within that specific octet block:

- **1 Bit Borrowed:** Custom Mask value = . 128 (Prefixes: /1, /9, /17, /25)
- **2 Bits Borrowed:** Custom Mask value = . 192 (Prefixes: /2, /10, /18, /26)
- **3 Bits Borrowed:** Custom Mask value = . 224 (Prefixes: /3, /11, /19, /27)
- **4 Bits Borrowed:** Custom Mask value = . 240 (Prefixes: /4, /12, /20, /28)
- **5 Bits Borrowed:** Custom Mask value = . 248 (Prefixes: /5, /13, /21, /29)
- **6 Bits Borrowed:** Custom Mask value = . 252 (Prefixes: /6, /14, /22, /30)
- **7 Bits Borrowed:** Custom Mask value = . 254 (Prefixes: /7, /15, /23, /31)
- **8 Bits Borrowed:** Custom Mask value = . 255 (Prefixes: /8, /16, /24, /32)